

Discretized, First-Order-Kinetics Simulation of the Catabolism Steps of Major Macronutrients of a Peanut in the Human Body

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1 Introduction

This paper presents a software simulation prototype of the catabolism of key peanut macronutrients within the human body. The simulation begins with chosen quantities of starch, proteins, and fats in the mouth. It concludes in the mitochondria, with final outputs of produced ATP or energy equivalents from the Tricarboxylic Acid (TCA) cycle and Electron Transport Chain (ETC). The simulation does not include major anabolic steps which influence the fate of peanut nutrients.

A video demo of the simulation is accessible at the following link:

https://drive.google.com/file/d/1iAvv22Su3pxp-vrJf5WH93K_mMK0f0L_/view?usp=sharing

To construct this simulation, successful enzymatic reaction rate models have been selected from the literature. Individual catabolic steps are intended to appear realistic. However, the project prioritized learning and qualitative demonstration over quantitative accuracy. As discussed below, strong assumptions made for the simulation model fail to capture the true complexities of metabolism.

Further sections address simulation construction, models used in each organ and metabolic process, assumptions, and limitations. Most of the theory which influenced simulation design and discussion came from graduate course lecture content (Brunton 2025). This content has been explicitly cited below only in instances where specific numerical values were used in calculations, but it served as the basis for the majority of the content design.

2 Simulation Construction

2.1 Micronutrient Inputs

To represent a peanut beginning ingestion in the mouth, quantities of carbohydrates, fats, and proteins corresponding to 50 grams of whole peanuts were selected. For simplicity, the model only considered the most prominent of each of these macronutrients in peanuts: starch, oleic acid (monounsaturated) (Settaluri et al. 2012), and α -arachin (Cui et al. 2023), respectively. All triglycerides were assumed to be triolein.

See Appendix I: Physical Constants for a complete list of sourced molar masses for these nutrients and other relevant physical constants used in the model.

2.2 Organ Pipeline Model

Table 1 shows a list of all considered organs/organelles, alongside each metabolic process within that organ and the processed substrates.

Table 1: Organs, Metabolic Agents, and Metabolites in the Peanut Metabolism Simulator

Organ	Metabolic Agent	Transported Metabolites	Internal Metabolites/Regulators
Mouth	Salivary amylase	Starch, maltose, triolein, arachin	–
Stomach	Gastric amylase, pepsin	Starch, maltose, arachin, polypeptide	–
Small Intestine	Pancreatic amylase, trypsin, lipase	Starch, maltose, polypeptide, peptide chain, triolein, diacylglycerol (DAG), monoacylglycerol (MAG), oleic acid	–
Small Intestine Lining	Maltose hydrolysis, peptidase, mixed micelle formation	Maltose, glucose, peptide chain, glutamic acid, bile salt, micelle	MAG, oleic acid
Enterocytes	Chylomicron production	Micelle, chylomicron	–
Pancreas	Insulin production	Insulin	–
Liver	Bile salt production	Bile salt	–
Blood	Lipoprotein lipase, glucose uptake regulator	Glucose, insulin, chylomicrons, oleic acid, glycerol, glutamic acid	Chylomicron remnant
Cytosol	Glycolysis, acyl-CoA synthase	Glucose, pyruvate, oleic acid, fatty acyl-CoA	G6P, F6P, F16BP, 3PG, PEP
Mitochondrial Matrix	α -Alanine amino transferase, TCA cycle, fatty acid oxidation	Pyruvate, glutamic acid, fatty acyl-CoA, alanine, α -ketoglutarate	GTP, acetyl-CoA
Mitochondrial Inner Membrane	Electron transport chain	NADH, FADH ₂ , ATP	–

Note: The term "organ" was applied to describe a general area, or set of areas, within the metabolic pipeline. Not all items categorized here, such as blood or the cytosol, meet the typical definition of an organ.

The organs in Table 1 were organized in a directional, pseudo-acyclic graph structure. In each simulation time step, an organ receives input metabolites from its predecessor (for example, the small intestine takes maltose from the stomach). After processing, the organ outputs relevant substrates downstream (i.e., the small intestine outputs glucose to the intestinal lining). This graph structure is demonstrated in Figure 1.

This graph-based pipeline is not fully acyclic because the blood acts as an input, output, and regulatory aid for other organs. In this model, for example, the pancreas outputs insulin to the bloodstream. However, it also must access the quantities (inputs) of glucose and amino acids within the blood to produce the correct amount of insulin (Morettini et al. 2022).

Each listed organ was treated as a fully isolated node which only receives inputs as discrete events. No continuous passage of substrates occurred in the model.

2.3 Declarative Model Organization

Within the code, the model was designed to have a declarative structure, such that functioning code also logically outlined the organization of the human body. Each organ was represented as an object that contained its metabolic agents and rate-limiting functions (Figure 2a). Each metabolic agent contained a list of its substrates and regulators (Figure 2b). To create the graph structure, organs were placed within nodes, alongside edges including output metabolites and their destinations (Figure 2c).

2.4 Simulation Dynamics

The simulation executed in discrete time steps of one minute. In simulation time, each time step took approximately 0.1 seconds. This step size was chosen because all reviewed publications of metabolism kinetics expressed rates in minutes.

In each time step, the above organ graph was traversed in a breadth-first search. In the first time step, initial peanut macronutrients (subsection 2.1) were input directly to the mouth. Then, the mouth executed kinetics for all contained metabolic agents (none) and placed destination organs for all output edges (only the stomach) into a queue. The next organ-node was removed from the queue, its metabolic agents executed, and destination organs added to the queue. This process iterated until all organs had been visited.

For all future time steps, the above process repeated, starting with the mouth. However, nutrients were only input into the mouth in the first time step, modeling a single mouthful of peanuts ingested all at once.

3 Explanation of Mathematical Models

From reviewed kinetics publications, first-order and pseudo-first-order (Chemistry LibreTexts 2013) exponential models were common. Others presented a form of the Michaelis-Menten equation, a popular representation for enzymatic kinetics (Srinivasan 2022). Ten such models were employed within the simulation. Most models were adopted assuming that the reaction would never be limited by the availability of the enzyme, but only the concentration of the substrate.

Not all reactions throughout the simulation used a tested model from the literature. Instead, some included naive approaches to facilitate demonstration of the end-to-end pipeline. Table 2 summarizes the substrates, enzymes, and kinetics equation type for each metabolic process. Below, the formulation of each process in the table is detailed.

All reaction models were solved numerically using a first-order, ordinary differential equations solver (see Appendix II: Software Design Details).

3.1 Mouth

The mouth was assumed to fully empty its contents into the stomach within a single, one-minute time step. The presence of the esophagus between the mouth and stomach was omitted.

3.1.1 Amylase

For simplicity, starch is treated as 100 percent amylose, which is converted to maltose by amylolysis. However, real starch metabolism would also include a pathway to catabolize contained amylopectin (Butterworth, Warren, et al. 2012). Real amylolysis would also produce maltotriose and α -limit dextrins (Roberts and Whelan 1960; Butterworth, Bajka, et al. 2022).

The mouth contains salivary amylase, and other types of amylase are included in organs below. All amylase reactions were modeled after the same kinetics for pancreatic amylase, a first-order exponential with a rate constant of 0.03 (or starch half-life of 23 minutes) (Butterworth, Warren, et al. 2012). Because a model of amylase action specifically on peanuts was not available, the results provided by Butterworth et al. (2012) for beans were selected. Peanuts and beans are in the same legume family and share similarities in macronutrients (Toomer 2018).

3.2 Stomach

3.2.1 Gastric Amylase

The same model presented in subsection 3.1.1 acted within the stomach on remaining starch passed from the mouth.

3.2.2 Pepsin

As no protein catabolism was considered within the mouth, all proteins (α -arachins) were passed directly to the stomach. Pepsin converted arachin into arachin polypeptides. As a proxy for this reaction, a model fitting the hydrolysis of milk casein by pepsin was used, with a first-order exponential rate constant of 0.08 (protein half-life of approximately 9 minutes) (Yang et al. 2022).

Effects on the reaction from pepsin denaturation were not considered. However, other models have considered this effect, particularly for increased denaturation at pH levels over 5.0 (Yang et al. 2022). Stomach acidity may approach this level after ingestion of a meal (Academic Press 2006).

Our model further assumed that pepsin was already active in the stomach upon food ingestion, ignoring preliminary activation of pepsinogen to pepsin by hydrochloric acid within the stomach.

3.2.3 Stomach Emptying Regulator

After nutrients in the stomach have undergone enzymatic reactions and formed into acid chyme, they are emptied into the duodenum of the small intestine. This step is tightly controlled via the pyloric sphincter at a potential rate of 2–4 kilocalories per minute (Kong and Singh 2008).

Our simulation used a power exponential function reviewed by Kong et al. (2008). This model resembled other first-order exponential models discussed but included a power constant to account for an initial lag period in the emptying of solid foods. With this model, 50 percent of solids were expected to be emptied in approximately 94 minutes, and completely emptied in 2 or more hours (Siegel et al. 1988; Kong and Singh 2008).

While this model did not differentiate between distinct macronutrients, as chyme passes to the small intestine in real metabolism, a striation of chyme would be expected, with emulsified fats on top. Our simulation did not account for such separation and instead assumed that all nutrients were distributed and emptied in equal proportion.

3.3 Small Intestine

3.3.1 Pancreatic Amylase

The model for pancreatic amylase matched the details of subsection 3.1.1.

3.3.2 Trypsin

To simplify, trypsin was the only enzyme considered in converting polypeptides within the small intestine into small peptide chains. However, a more robust model should also consider the presence of chymotrypsin and carboxypeptidase in this step.

Our model based the catabolic rate of polypeptides on findings by Schwert and Eisenberg (1949) of the action of trypsin on benzoyl-L-argininamide. While the authors noted that the reaction theoretically requires higher-order modeling due to varying levels of competitive inhibition depending on substrate concentration, experimental results followed a zero-order (linear) hydrolysis rate of 1.1×10^{-3} millimoles per minute of polypeptides (Schwert and Eisenberg 1949). This same zero-order model was selected for simulation.

3.3.3 Lipase

Lipases which catalyze the conversion of triglycerides to di- and mono-glycerides are present in the mouth, stomach, small intestine, and intestinal lining. Because most lipolysis occurs within the small intestine after emulsification via bile salts (not available to the stomach or mouth), only lipase within the small intestine (pancreatic lipase) was considered.

The selected model of pancreatic lipase presented distinct first-order reactions for the hydrolysis of triolein to 1,2-diolein, and then 1,2-diolein to 2-monolein. Each reaction was dependent on the initial concentration of triolein and rate constants for both substeps. Diglycerides were catabolized nearly three times faster than triglycerides (Lykidis, Mougios, and Arzoglou 1995).

3.4 Small Intestine Lining

3.4.1 Maltose Hydrolysis

Maltose resulting from amylolysis earlier in the organ pipeline was hydrolyzed to glucose. This reaction is catalyzed by maltase. A Michaelis-Menten equation model was selected, whose numerical parameters were derived from experiments with rat small jejunums (Gromova, Fetissov, and Gruzdkov 2021).

To apply this model to humans, a typical human small intestine length (Fish, Shumway, and Burns 2025) and fed intestinal fluid volume (Schiller et al. 2005) replaced the values used for rats.

3.4.2 Peptidase

Peptidase catalyzes the hydrolysis reaction converting peptide chains into individual amino acids (all assumed to be glutamic acid).

The rate for this step is not sourced. For demonstration, it is assumed to be first-order, catabolizing 20 percent of existing small peptide chains during each time step.

3.4.3 Mixed Micelle Formation

To enter the enterocytes, monoglycerides and free fatty acids within the small intestinal lining first form emulsified mixed micelles with bile salts.

Kinetics for this process were not sourced. A naive model was implemented in which a single mole of monolein, two moles of oleic acid, and one mole of bile coalesce to form a mole of micelle. The lowest-quantity substrate was selected as the limiting reactant in each time step, and the reaction consumed 50 percent of the limiting reactant alongside corresponding proportions of both other substrates.

3.4.4 Chylomicron Production

After passing into the enterocytes via mixed micelles, glycerides and fatty acids form into chylomicrons, which are water-soluble and hence may enter the bloodstream.

Enterocytes were modeled as a distinct organ in the simulation to emphasize this step. For simplicity, 50 percent of micelles from the intestinal lining per time step were assumed to eventually produce chylomicrons, with all intermediate steps omitted.

3.5 Liver

3.5.1 Bile Salt Production

Bile salt was modeled to be injected directly to the small intestine from the liver. However, in reality, bile would first be stored within the gallbladder and later delivered via the bile duct. Our modeled liver produced and transported 1 mmol of bile salt per time step.

3.6 Pancreas

3.6.1 Insulin Production

Insulin production by the pancreas, and secretion to the bloodstream, relied on a model with both plasma glucose and plasma amino acids as regulators. To determine the needed level of insulin in a time step, the model assigned linear coefficients to current levels of glucose, amino acids, and insulin. Those coefficients were applied to the differences between current levels and assumed base levels, such that a higher change from base levels corresponded to a higher change in insulin (Morettini et al. 2022).

We adopted this model by communicating calculated values of glucose and glutamic acid within the simulated bloodstream to the pancreas node, which were used as parameters in the provided linear equation.

3.7 Bloodstream

3.7.1 Insulin Degradation

The same model used in subsubsection 3.6.1 accounted for insulin degradation within the blood via first-order feedback. Assuming a steady-state average insulin level I_b of approximately $7.5 \mu\text{U/mL}$ and a rate coefficient k_I of 0.14, insulin in the next time step was denatured by (Morettini et al. 2022):

$$Insulin(t) = k_I * (Insulin(t - 1) - I_b)$$

In this manner, insulin was expected to naturally degrade to its base level in steady state. The specific kinetics of the insulin-degrading enzyme, which contribute to an expected insulin half-life of 4–6 minutes (Academic Press 2009), were not included in this approach.

3.7.2 Glucose Uptake Regulator

Insulin levels in the blood affect the rate at which glucose is absorbed into cells. Passage of glucose from the blood across the cell membrane was implemented as a one-way edge between the blood and cytosol nodes. Glucose uptake regulation was implemented as a rate limiter for this edge.

Estimated parameters were selected from a Michaelis-Menten equation of glucose uptake, fit to glucose and insulin readings of human adults. This equation modeled the maximum glucose uptake per unit time as a Hill function (Goutelle et al. 2008) of insulin concentration at the active site (Bizzotto et al. 2016).

3.8 Within Cells

Distinct areas of cells, including the cytosol, mitochondrial matrix, and inner mitochondrial membrane, were modeled as though the volume of participating cells of the body were combined into a single cell, with equal distribution of nutrients from the blood to all areas. This volume was visualized as single, pseudo-organ nodes for each area. Types of cells were not differentiated, and a total count of 3.00×10^{13} cells participating in glycolysis and energy generation was assumed (Hatton et al. 2023).

3.8.1 Glycolysis - Cytosol

Glucose which enters the cell cytosol undergoes glycolysis with the final output of pyruvate. The simulation assumed only aerobic conditions and, hence, that zero pyruvate was ever later reduced to lactic acid.

Further, glycolysis and other catabolic processes within the cells treated adenosine triphosphate (ATP) levels as effectively unlimited. Their availability was not considered for substrate inputs nor outputs (excluding the ETC).

The selected model considered seven different intermediate substrates, in order of appearance in glycolysis: glucose, glucose-6-phosphate (G6P), fructose-6-phosphate (F6P), fructose-1,6-bisphosphate (F16BP), 3-phosphoglycerate (3PG), phosphoenolpyruvate (PEP), and pyruvate. The rate of change of each was modeled with Michaelis-Menten and Hill kinetics as a function of other intermediates (Rehberg, Ritter, and Reichl 2014).

Table 3 shows highly simplified relationships between each of these substrates. While the table omits feedback effects on particular glycolytic enzymes, expected examples of feed-forward and backward regulation are present. For example, F16BP increases the later production of PEP from 3PG by allosterically activating pyruvate kinase (Rehberg, Ritter, and Reichl 2014). As a backwards-feedback example, G6P's inhibition of hexokinase is encoded in Table 3 as "self-regulation."

Our simulation lacked key glycolysis regulation steps due to the omission of non-substrate regulators. In particular, no enzymes were regulated by the presence of energy-yielding molecules such as ATP, adenosine diphosphate (ADP), and adenosine monophosphate (AMP).

3.8.2 Fatty Acid Oxidation – Mitochondrial Matrix

The reactions below within the mitochondrial matrix are implemented naively in that inputs from previous steps were assumed to be converted directly into energy-yielding molecules. However, net quantities of energy-yielding molecules output were expected to be realistic. To retain a pool of cycle inputs for visualization, only 75 percent were processed in each time step for fatty acid oxidation, conversion of amino acids to alanine, and the TCA cycle.

1 mole of Fatty acyl-CoA produced a mole of acetyl-CoA based on the length of the fatty acid carbon chain:

$$Acetyl - CoA = N_C / 2 - 1$$

in addition to 1 mole NADH and 1 mole FADH₂ (Brunton 2025). As oleic acid has a carbon length of 18 (Settaluri et al. 2012; PubChem n.d.[d]), each step produced 8 moles of Acetyl-CoA per mole of oleic acid.

3.8.3 Amino Acid Catabolism - Mitochondrial Matrix

Glutamic acid and pyruvate react to form α -ketoglutarate and alanine when catalyzed by α -alanine aminotransferase (Lenartowicz and Wojtczak 1988). α -ketoglutarate can then enter the TCA cycle (subsubsection 3.8.4). Our model assumed that glutamic acid would always be the limiting substrate of this reaction, such that, in each time step, 75 percent of contained glutamic acid was consumed along with an equal number of moles of pyruvate.

3.8.4 TCA Cycle - Mitochondrial Matrix

Each mole of acetyl-CoA and pyruvate produced 3 moles NADH, 1 mole FADH_2 , and 1 mole guanosine triphosphate (GTP). α -ketoglutarate produced 2 moles NADH, 1 mole FADH_2 , and 1 mole GTP (Brunton 2025). The mitochondrial matrix included two processes with the same input substrate—both amino acid catabolism and the TCA cycle required pyruvate. Distribution of pyruvate to these two processes was implemented via a simple, ordered list: first, all available glutamate was converted; then, all remaining pyruvate was passed to the numerical solver for the TCA.

3.8.5 ETC - Mitochondrial Inner Membrane

The ETC model assumed that 2 moles ATP were produced per mole NADH and 1.5 moles per mole FADH_2 . The maximum production rate was 100 ATP molecules per minute per ATP-synthase (Brunton 2025). Total production considered the total number of active synthases as a product of synthases per mitochondria, mitochondria per cell, and total cells (Appendix I: Physical Constants).

4 Discussion

4.1 Strengths

This simulation presented a unified set of researched reaction kinetics. While such kinetics of major metabolic steps are accessible in the literature, relatively little has been published combining many together to demonstrate larger-scale effects in the body.

This work closely coupled demonstrative code with an intuitive organization of metabolism and scientific sources. Based on a limited search, such design and prioritization of code accessibility appear to be novel in biochemical kinetics research.

4.2 Limitations

At the time of writing, this simulation is a prototype intended to demonstrate knowledge at an introductory graduate level. It included steps which are known to be based on incorrect assumptions and omitted important steps, particularly in regulation. As the software had not been part of a rigorous review, the code likely contained further defects.

Modeling for this simulation was not based on a thorough literature review. High-level model flow was taken from course lecture content (Brunton 2025), and more specific sources for individual reactions were selected on their strengths as easily accessible and relevant parameterized models. Projected accuracy within the overall metabolic pipeline was not the highest priority.

4.3 Future work

The possible improvements noted above are strong candidates to improve the robustness of the presented simulation. To remove overly strong assumptions, a literature review should be conducted to better select models for each interaction. More recently researched models should be added, including realistic models of the transfer of metabolites between organs and organelles (not considered in the current simulation).

The code should be reviewed and improved for maintainability by multiple investigators. It can then be considered a tool to continue mapping intuitive physical body structure to functioning code which demonstrates the current state of the literature. Such a tool will be useful to unify metabolism research and identify gaps.

No anabolic processes have yet been included, such as the transportation of glycerol from triglyceride catabolism to the liver for either gluconeogenesis or fatty acid reconstruction. Addition of these processes would augment the demonstrative capability of this simulation as a learning tool. Further, the developed codebase is equipped to support such new transports and processes with minimal time investment.

5 Appendix I: Physical Constants

Table 4: Physical Constants Used in Metabolism Simulation

Constant	Value	Units	Metabolic Process(es)	Source
Fundamental Constants				
Avogadro Number	6.022×10^{23}	molecules/mol	Electron Transport Chain	–
Human Body Parameters				
Average Human Body Surface Area	1.79	m ²	Glucose Uptake Regulation	(Sacco et al. 2010)
Blood Parameters				
Total Blood Volume	5.0	L	Glycolysis, Insulin, Blood Init	(R. Sharma and S. Sharma 2025)
Basal Glucose (<i>mg/dL</i>)	90	mg/dL	Insulin Production	(Morettini et al. 2022)
Basal Insulin	7.5	$\mu\text{U/mL}$	Insulin Production, Blood Init	(Schrank et al. 2024)
Basal BCAA	475	$\mu\text{mol/L}$	Insulin Production, Blood Init	–
Cell Parameters				
Total Cell Count	3.00×10^{13}	cells	Glycolysis, ETC	(Hatton et al. 2023)
Single Cell Volume	3.04×10^{-12}	L/cell	Glycolysis	(Rehberg, Ritter, and Reichl 2014)
Mitochondria per Cell	1000	–	Electron Transport Chain	(Cole 2016)

Table 4 continued from previous page

Constant	Value	Units	Metabolic Process(es)	Source
Active ATP Synthase per Mitochondria	1000	–	Electron Transport Chain	(Fahimi and Matta 2021)
Molar Masses				
Amylose	200,000	g/mol	Salivary/Pancreatic Amylase, Input	(ScienceDirect n.d.)
Maltose	342.30	g/mol	Salivary/Pancreatic Amylase	(PubChem n.d.[c])
Glucose	180.16	g/mol	Insulin Production	(PubChem n.d.[a])
Arachin	490,000	g/mol	Pepsin, Input	(Basha and Pancholy 1981)
Arachin Polypeptide	68,000	g/mol	Pepsin, Trypsin	(Basha and Pancholy 1981)
Small Peptide	680	g/mol	Trypsin, Peptidase	Assumed 1% of polypeptide
Glutamic Acid	147.134	g/mol	Peptidase	(PubChem n.d.[b])
Triolein	885.4	g/mol	Input	(PubChem n.d.[e])
Fatty Acid Properties				
Oleic Acid (chain length)	18	carbons	Fatty Acid β -Oxidation	(Settaluri et al. 2012; PubChem n.d.[d])
Maltose Hydrolysis Parameters				
Lumen Volume (L)	0.054	L	Membrane Maltose Hydrolysis	(Schiller et al. 2005)

Continued on next page

6 Appendix II: Software Design Details

The simulation data modeling was implemented using the Kotlin Multiplatform (Kotlin n.d.[b]) toolset, popular for developing applications which compile for multiple runtimes. The simulation visualization was created with the corresponding UI framework, Kotlin Compose (Kotlin n.d.[a]). The simulation was built and packaged using Gradle (Gradle 2025), developed using the IntelliJ IDEA: Ultimate Edition (JetBrains n.d.), and tested on a 2024 Apple Mac mini with an Apple M4 CPU and 24 gigabytes of RAM. While the simulation application has been made available for both macOS and Windows distributions, it has not yet been tested on Windows.

Kinetics were numerically solved using a first-order differential equations solver from the Apache Maths Commons Java library (Apache Commons n.d.[b]). It was configured to use the Dormand-Prince 853 integrator with default parameters (Apache Commons n.d.[a]). Optimization of these parameters would likely improve the accuracy of simulation performance. This is a candidate for future work.

All code and build artifacts from this work are publicly available on GitHub:
<https://github.com/jsharp4/PeanutsMetabolismSimulator>

Claude Code was used via a command-line terminal during project development. See Appendix III: A.I. Usage.

7 Appendix III: A.I. Usage

Commercially available A.I. tools were used to assist with the construction of this simulation and for formatting aspects of this paper. All suggested output was verified by this author. All ideas within the paper and in simulation data design are cited or the original work of this author.

Claude Code was used for code refactoring, assistance with triaging issues within the application, formatting Kotlin Compose visuals, and constructing drafts of tables from information already manually embedded in the code.

Google AI Studio was used to extract numerical values, units, equations, and abbreviations from publications into formats usable within Kotlin code, to improve the formatting of LaTeX markup, and to search for the location of known content within publications. It was also used for code debugging assistance and written grammar/proofreading.

Archives of all conversations with Claude Code and AI Studio are available at:

<https://github.com/jsharp4/PeanutsMetabolismSimulator/tree/final-video-version/ai-assistance-archives>

8 Appendix IV: Illustration Sources

Table 5: Sources for Simulation Diagram Components and Illustrations

Component / Anatomical Feature	Source
Eukaryote cell	(Britannica 2025a)
Mitochondrion, inner membrane	(Britannica 2025d)
Mouth	(Britannica 2025e)
Stomach	(Britannica 2025g)
Small intestine, lining, enterocyte	(Britannica 2025f)
Circulatory system	(Britannica 2025b)
Liver	(Britannica 2025c)
Pancreas	(Pereira and Sánchez 2018)

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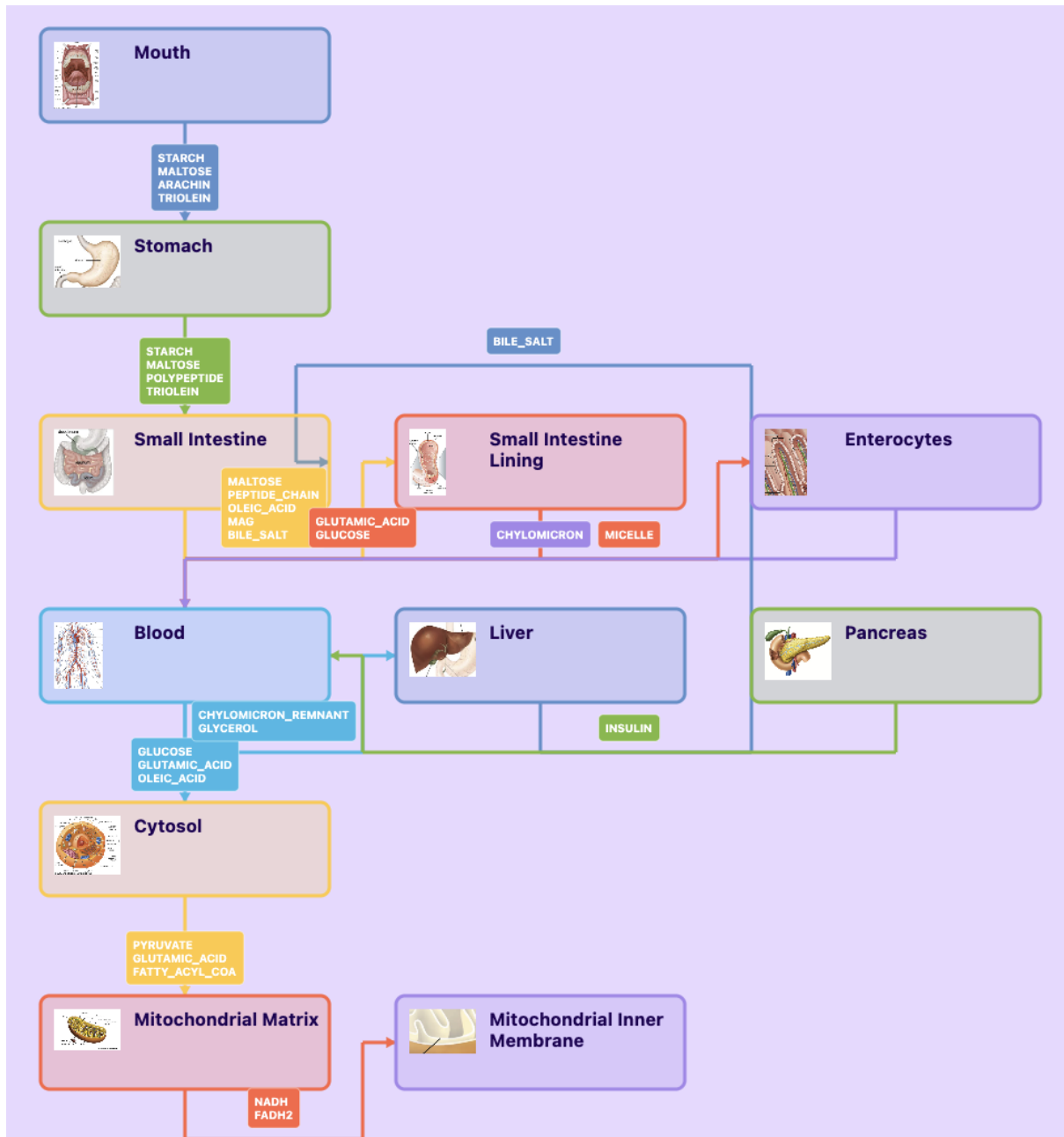


Figure 1: Directional graph view of flow of metabolites between organs, including key regulators and metabolites passed through the pipeline

```

class MitochondrialMatrix(): Organ(
    metabolicProcesses = listOf(
        AlanineAminoTransferase(),
        TCACycle(),
        FattyAcidOxidation()
    )
)

```

(a) organelle with metabolic processes

```

class AlanineAminoTransferase: MetabolicProcess(
    ode = AminoTransferaseODE(),
    metaboliteTypes = listOf(
        MetaboliteType.GLUTAMIC_ACID,
        MetaboliteType.PYRUVATE,
        MetaboliteType.KETOGLUTARATE,
        MetaboliteType.ALANINE
    )
)

```

(b) metabolic process with substrates

```

val mitochondrialMatrixNode = OrganNode(
    organ = mitochondrialMatrix,
    ...edges = DirectionalOrganEdge(
        destination = mitochondrialMembraneNode,
        ...metabolitesToSend = QuantifiedMetabolite( type = MetaboliteType.NADH, percentageOfOutput = 100.0),
        QuantifiedMetabolite( type = MetaboliteType.FADH2, percentageOfOutput = 100.0)
    )
)

```

(c) edge connecting outputs of organelle to another organelle

Figure 2: Example declarative instantiations of physical metabolic organization within code, using the mitochondria (a), the action of α -alanine aminotransferase (b), and the passage of NADH and FADH₂ to the inner mitochondrial membrane (c)

Table 2: Metabolic Process, Metabolites, and Kinetic Equations

Metabolizer	Metabolites	Equation Type	Source
Salivary amylase	Starch $\rightarrow$ Maltose	First-order exponential	(Butterworth, Warren, et al. 2012)
Gastric amylase	Starch $\rightarrow$ Maltose	First-order exponential	(Butterworth, Warren, et al. 2012)
Pepsin	Arachin $\rightarrow$ Polypeptide	First-order exponential	(Yang et al. 2022)
Stomach emptying	Starch, maltose, polypeptide, triolein	Exponential power-law	(Siegel et al. 1988; Kong and Singh 2008)
Pancreatic amylase	Starch $\rightarrow$ Maltose	First-order exponential	(Butterworth, Warren, et al. 2012)
Trypsin	Polypeptide $\rightarrow$ Peptide chain	Zero-order	(Schwert and Eisenberg 1949)
Lipase	Triolein $\rightarrow$ DAG $\rightarrow$ MAG $\rightarrow$ Oleic acid	Sequential first-order	(Lykidis, Mougios, and Arzoglou 1995)
Membrane maltose hydrolysis	Maltose $\rightarrow$ Glucose	Michaelis-Menten	(Gromova, Fetissov, and Gruzdkov 2021)
Peptidase	Peptide chain $\rightarrow$ Glutamic acid	First-order	Assumed
Mixed micelle former	MAG + Oleic acid + Bile salt $\rightarrow$ Micelle	Limiting reactant	Assumed
Chylomicron producer	Micelle $\rightarrow$ Chylomicron	First-order	Assumed
Bile salt producer	$\rightarrow$ Bile salt	Zero-order	Assumed
Insulin production	Glucose + Glutamic acid $\rightarrow$ Insulin	Linear with feedback	(Morettini et al. 2022)
Lipoprotein lipase	Chylomicron $\rightarrow$ Chylomicron remnant + Oleic acid + Glycerol	Zero-order	Assumed
Glucose uptake regulator	Glucose, insulin	Michaelis-Menten + Hill	(Bizzotto et al. 2016)
Insulin breakdown	Insulin $\rightarrow$	First-order decay	(Morettini et al. 2022)
Glycolysis	Glucose $\rightarrow$ G6P $\rightarrow$ F6P $\rightarrow$ F16BP $\rightarrow$ 3PG $\rightarrow$ PEP $\rightarrow$ Pyruvate	Michaelis-Menten + Hill	(Rehberg, Ritter, and Reichl 2014)
Acyl-CoA synthase	Oleic acid $\rightarrow$ Fatty acyl-CoA	First-order	Assumed
Alanine aminotransferase	Glutamic acid + Pyruvate $\leftrightarrow$ α -ketoglutarate + Alanine	First-order	Assumed
TCA cycle	Pyruvate, acetyl-CoA, α -ketoglutarate $\rightarrow$ NADH, FADH ₂ , GTP	First-order	(Brunton 2025)
Fatty acid oxidation	Fatty acyl-CoA $\rightarrow$ Acetyl-CoA + NADH + FADH ₂	First-order	(Brunton 2025)
Electron transport chain	NADH + FADH ₂ $\rightarrow$ ATP	Michaelis-Menten	(Brunton 2025)

Table 3: Glycolysis ODE Metabolite Dependencies

Metabolite	Dependency	Effect
GLC (Intracellular Glucose)	GLC	$\pm$ (self-limiting)
G6P	GLC	+
	G6P	− (self-limiting)
	F6P	+
F6P	G6P	+
	F6P	− (self-limiting)
F16BP	F6P	+
	F16BP	− (self-limiting)
3PG	F16BP	+
	3PG	− (self-limiting)
	PEP	+
PEP	3PG	+
	PEP	− (self-limiting)
	F16BP	−
Pyruvate	PEP	+
	F16BP	+